

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q18-1094

Candelaria 138 kV
February 01, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1094. The Interconnection Customer proposes a 57.8 MW solar photovoltaic facility interconnecting at the Candelaria 138 kV substation in the Virginia area. The request was received on October 31, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 407 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 19 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$15,097,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Oak Creek to Carmel 146Th St 138.0 kV	Reach to Candelaria 138.0 kV line	146.6	200	6.4
Ceiba Baja 7012 to Carmel 146Th St 138.0 kV	Candelaria to Reach 138.0 kV line	145.4	204	6.4
Tangy to Ceiba Baja 7012 138.0 kV L	Reach to Candelaria 138.0 kV	134.1	229	6.4
Carson Ice Cogen to Alderwood 138.0 kV	Reach to Candelaria 138.0 kV	126.3	318	12.0
Bluff City to Reach 138.0 kV	Scottsville to Reach 138.0 kV L	121.0	397	49.2
Tangy to Barksdale 138.0 kV	Reach Substation to Candelaria 138.0 kV L	120.9	214	6.1
Amber to Bremono Bluff 138.0 kV	Lake Lure to Bluff City 138.0 kV	119.4	249	11.3
Grassmere to Red Hill 138.0 kV	Reach to Candelaria 138.0 kV	117.1	295	9.7
Scottsville to Reach 138.0 kV	Bluff City to Reach 138.0 kV	114.6	345	39.1
Walworth to Amber 138.0 kV line	Bluff City to Lake Lure 138.0 kV	114.2	214	12.1
Coal Mtn to Grassmere 138.0 kV	Reach to Candelaria 138.0 kV	114.0	329	9.7
Coal Mtn to Central Austin 138.0 kV	Reach to Candelaria 138.0 kV	113.3	338	9.7
CentraAustin to Waynesboro 138.0 kV L	Reach to Candelaria 138.0 kV	112.5	348	9.7
Waynesboro to Alderwood 138.0 kV	Reach to Candelaria 138.0 kV	110.6	377	9.7
Charlottesville to Mohican 138.0 kV	Red III 345/138kV Transformer Bank	105.1	168	6.9
Reach to Candelaria 138.0 kV	Bear to Clover 345.0 kV Line	104.5	631	56.0

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Grove St to Fluvanna P.S 138.0 kV Line	Red Hill 345/138kV XFMR	103.1	345	10.6
Scottsville to Rock Island Dam 138.0 kV	Bluff City to Reach 138.0 kV	102.9	235	39.1
Yankee Doodle to Rock Island Dam 138.0 kV	Reach to Bluff City 138.0 kV L	102.0	217	39.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Crmel 146Th St to Oak Crk 138.0 kV Line	Rebuild line with higher-capacity conductor	\$8,777,000	4% (\$377,000)
Carmel 146Th St to Ceiba Baja 7012 138.0 kV Line	Rebuild line with higher-capacity conductor	\$13,711,000	4% (\$588,000)
Tangy to Ceiba Baja 7012 138.0 kV L	Rebuild line with higher-capacity conductor	\$4,540,000	4% (\$195,000)
Carson Ice Cogen to Alderwood 138.0 kV	Rebuild line with higher-capacity conductor	\$4,434,000	10% (\$421,000)
Bluff City to Reach 138.0 kV	Rebuild line with higher-capacity conductor	\$26,826,000	12% (\$3,273,000)
Tangy to Barksdale 138.0 kV	Rebuild line with higher-capacity conductor	\$33,321,000	12% (\$3,908,000)
Amber to Bremo Bluff 138.0 kV	Rebuild line with higher-capacity conductor	\$17,046,000	6% (\$1,050,000)
Grassmere to Red Hill 138.0 kV	Rebuild line with higher-capacity conductor	\$3,631,000	4% (\$156,000)
Scottsville to Reach 138.0 kV	Rebuild line with higher-capacity conductor	\$8,662,000	12% (\$1,052,000)
Walworth to Amber 138.0 kV line	Rebuild line with higher-capacity conductor	\$19,357,000	12% (\$2,370,000)
Coal Mtn to Grassmere 138.0 kV	Rebuild line with higher-capacity conductor	\$3,703,000	4% (\$159,000)
Coal Mtn to Central Austin 138.0 kV Line	Rebuild line with higher-capacity conductor	\$3,766,000	4% (\$162,000)
Central Austin Substation to Waynesboro 138.0 kV	Rebuild line with higher-capacity conductor	\$3,111,000	4% (\$134,000)
Waynesboro to Alderwood 138.0 kV	Rebuild line with higher-capacity conductor	\$6,148,000	4% (\$265,000)
Charlottesville to Mohican 138.0 kV	Reconductor line	\$1,252,000	4% (\$46,000)
Candelaria to Reach 138.0 kV	Reconductor line	\$3,377,000	10% (\$336,000)
Grove St to Fluvanna P.S 138.0 kV Line	Reconductor line	\$387,000	2% (\$10,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Rock Island Dam Substation to Scottsville 138.0 kV	Reconductor line	\$3,594,000	10% (\$355,000)
Yankee Doodle to Rock Island Dam 138.0 kV Line	Reconductor line	\$2,426,000	10% (\$240,000)
	Total allocated to project		\$15,097,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.